

A Mini Project Report

On

SEQUENCE LABELING USING GENERATIVE

ADVERSARIAL NETWORKS

Submitted in the Partial Fulfillment of the Requirements

For the Award of the Degree of

Bachelor of Technology

In

CSE(Artificial Intelligence and Machine Learning)

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CERTIFICATE

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CANDIDATE’S DECLARATION

We Hereby Certify that the Work Which is Being Presented in the Mini Project Entitled “**SEQUENCE LABELING USING GENERATIVE ADVERSARIAL NETWORKS**” in Partial Fulfillment of the Requirements For the Award of Degree of Bachelor of Technology and Submitted in the Department of CSE(Artificial Intelligence and Machine Learning), B V Raju Institute of Technology, Narsapur, is an Authentic Record of Our Own Work Carried Out During the Period From December 2024 to May 2025, Under the Supervision of Mrs. B. Lavanya, Assistant Professor and Ms. Srilakshmi V, Assistant Professor.

The Work Presented in this Mini Project Report Has Not Been Submitted By Us For the Award of Any Other Degree of This or Any Other Institute/University.

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ABSTRACT

Sequence labeling plays a central role in a variety of natural language processing (NLP) applications, including Named Entity Recognition (NER), Part-of-Speech (POS) tagging, and Chunking. While pretrained language models have greatly improved the performance of these tasks, they typically require large quantities of annotated data and computational resources to train effectively. This work presents a lightweight, adversarial trained sequence labeling framework that aims to overcome these limitations by introducing an extra training signal that encourages the model to produce more consistent and accurate label sequences.

The proposed method uses two main components: a tagger and a checker. The tagger reads a sentence and predicts a label for each word (such as a part of speech or named entity), using a small, efficient language model followed by simple layers that refine the predictions. The checker acts like a reviewer—it looks at both the predicted and correct label sequences and learns to tell the difference. To enable seamless training of the checker, the right labels are somewhat altered to look more like probabilities than strict one-hot values. For the reviewer, this makes learning more flexible and instructive. In addition to obtaining correct answers, the tagger aims to produce realistic label sequences that trick the checker during training. Especially in challenging circumstances, this extra restriction helps the tagger avoid making inaccurate or inconsistent predictions. The result is a model that learns and gets better even with scant training data.

We conduct experiments with other English sequence labeling tasks, including CoNLL-2003 (NER), Universal Dependencies (POS), and CoNLL-2000 (Chunking). Results show that this approach uses significantly less computation while delivering high F1 scores, demonstrating its usefulness in low-resource or time-critical environments.

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LIST OF ABBREVIATIONS

NLP	Natural Language Processing
AI	Artificial Intelligence
NER	Named Entity Recognition
POS	Part-of-Speech
GAN	Generative Adversarial Network
BERT	Bidirectional Encoder Representations from Transformers
CRF	Conditional Random Field
RNN	Recurrent Neural Network
OOV	Out-of-Vocabulary
LLM	Large Language Model
TPU	Tensor Processing Unit
GPU	Graphics Processing Unit
VAT	Virtual Adversarial Training
BIO	Beginning-Inside-Outside (tagging scheme)
NP/VP/PP	Noun Phrase / Verb Phrase / Prepositional Phrase

CHAPTER 1

INTRODUCTION

1.1. INTRODUCTION TO TOPIC

Sequence labeling is one of the most important and well-studied tasks in Natural Language Processing (NLP). In sequence labeling, an input sequence receives a label, or a category, for each element in the sequence. The elements in the sequence are typically words or tokens in a sentence. Some of the most well-known applications of sequence labeling are Named Entity Recognition (NER), where people, organizations and locations are identified, Part-of-Speech (PoS) tagging where grammatical roles are assigned to words, and Syntactic Chunking where words are grouped into meaningful phrases like noun phrases, verb phrases, etc.



Figure 1.1 Sequence Labelling

These tasks lay the foundation for advanced NLP systems which are key aspects of several applications such as machine translation, text summarization, question answering and conversation. Good and reliable sequence labeling means that these more sophisticated tasks can reasonably and accurately understand and manipulate human language.

Deep learning methods have greatly advanced the state-of-the-art in sequence labeling models over the years. Architectures such as Recurrent Neural Networks (RNNs), and more recently Transformers, have improved the model's ability to learn the contextual meaning of words within a sentence. Also, pretrained language models improve the model's ability to learn more nuanced patterns of linguistics.

Also, despite these advances, a number of major limits still exist:

- Dependence on large, labeled datasets.
- Dealing with rare or unseen tokens.
- Cross-domain and robustness limits.

These limits may remind us of the current need for sequence labeling systems to be not only accurate and efficient, but also generalizable and robust across several language phenomena and usage contexts.

1.2. MOTIVATION

Modern natural language processing (NLP) systems frequently rely on engineered features, domain-specific rules, and large annotated datasets for maximal performance - in many real-world domains - including low-resource languages, new and changing contexts, or noisy forms such as social media - this is hard or impossible. Furthermore, traditional sequence labeling models learn to memorize surface-level patterns as opposed to learning about actual language, leading to a number of fundamental issues:

- Poor generalization across tasks and domains when encountering novel sentence structures or contextual clues.
- Failure to handle Out-of-Vocabulary (OOV) words that limits performance on unseen or rare tokens.
- Reduced accuracy when applied to informal and unstructured or noisy text such as user-generated content and informal speech.

This project is motivated by the quest to advance the performance of sequence labeling models to be more robust, efficient, and adaptable. While there is reliance upon large-scale supervised training, the main focus of this approach is on building efficient, lightweight transformer models (TinyBERT) with a softmax classification head. The intention is to ensure we exhibit a performance increase in core NLP tasks—Named Entity Recognition, Part-of-Speech tagging, and Chunking—while still avoiding computational burden and maintaining the ability to scale up for use in practical real-world scenarios.

1.3. SCOPE

The objective of this project is to design, develop, and test a lightweight GAN-based sequence labeling architecture specifically designed for the

core NLP tasks in English: Named Entity Recognition (NER), Part-of-Speech (PoS) Tagging, and Chunking. The proposed architecture takes its inspiration from the recent advances in adversarial learning applied to NLP. The proposed architecture utilizes lightweight architecture that replaces large models like BERT and complex decoders like CRF, while achieving comparable performance.

Key Components:

- **Data Collection:** We utilize three widely adopted English Datasets
 - CoNLL-2003 for Named Entity Recognition.
 - CoNLL-2000 for Syntactic Chunking.
 - Brown Corpus for PoS Tagging.
- **Model Development:**
 - **Generator:**
 - Architecture: TinyBERT + softmax-based sequence labeling head.
 - Function: Predicts label distributions for each token in the input sequence..
 - **Discriminator:**
 - Architecture: TinyBERT encoder + binary classifier head.
 - Function: Input tokens and label sequences (either real or generated labels) are taken as input to classify at the token level whether the labels are real or generated.
 - **Loss Functions:**
 - CrossEntropy loss for the generator (represented in the supervised learning component).
 - Adversarial loss (binary cross-entropy) from the discriminator to provide feedback on label quality.
 - **Training Pipeline:**
 - Implements a dual-stream adversarial training loop.
 - The generator produces predicted label sequences.

- The discriminator evaluates those predictions against true labels.
- The generator is updated on the basis of label correctness and feedback from the discriminator.
- Training incorporates subword alignment, masking, and efficient batching with PyTorch and HuggingFace.
- **Evaluation Metrics:**
 - Token-level F1 score, precision, recall, and accuracy.
 - Per-entity evaluation (note the rare ones such as I-MISC, I-ORG).
 - OOV (out-of-vocabulary) recall and confusion matrices for evaluation analysis.

1.4. PROBLEM STATEMENT

Sequence labeling is a basic task of Natural Language Processing (NLP) with important applications such as Named Entity Recognition (NER), Part-of-Speech (PoS) tagging, and Syntactic Chunking. While transformer-based architectures have significantly improved the state-of-the-art, many of the available solutions rely on large and complex models that make them impractical in real-world scenarios in which computing resources may be strictly limited.

This project presents a framework based on Generative Adversarial Networks (GANs) for sequence labeling that is both small, fast, and accurate. We address the following top-level issues related to existing models:

- Discrete sequence prediction in GANs.
- Defining a discriminator that produces more finely granular feedback.
- Achieving a balance between adversarial loss and supervised loss during training.

1.5. OBJECTIVES

- Create and deploy a GAN-based architecture that was designed specifically for sequence labeling tasks that involves a small compact transformer model and token-level feedback.
- Evolve a generator and discriminator architecture which uses both A) the context of input language and B) deriving adversarial improvements to the generator architecture, creating approaches for supervised labeling accuracy.
- Apply the architecture to core sequence labeling tasks, including:
 - Named Entity Recognition (NER)
 - Syntactic Chunking
 - Part-of-Speech (PoS) Tagging
- Train and evaluate architecture using benchmark datasets to commonly accepted English datasets, such as CoNLL-2003, CoNLL-2000, and the Brown Corpus.
- Evaluate the validity of standard evaluation metrics, including accuracy, precision, recall, F1-score, and detailed entity-level analysis.
- Compare the proposed model against traditional baselines for sequence labeling tasks, measuring effectiveness against another structured prediction approach.
- Conduct error analysis to consider the behaviors of the model focusing on instances where correlations exist, i.e. instances of rare, ambiguous, or difficult-to-predict sequences.
- Consider other avenues for future development of architecture (e.g. better balancing of loss function, domain adaptation, potential collaborative or auxiliary linguistic features).

1.6. ORGANIZATION OF THE REPORT

This mini project report is organized into five well-structured chapters, each of which covers a critical aspect of the work on “*Sequence Labeling Using Generative Adversarial Networks*.” The structure is designed to

offer a logical flow from problem identification to implementation, evaluation, and future directions.

Chapter 1: Introduction

- Introduces the domain of sequence labelling and its relevance in NLP.
- Explains foundational tasks such as NER, POS tagging, and Chunking.
- Identifies the limitations of traditional models and lists the need for a more efficient and robust solution.
- Clearly states the problem statement, objectives, and scope of the project.

Chapter 2: Review of Literature

- Provides an in-depth survey of recent and relevant research works in sequence labelling, adversarial learning, and lightweight transformers.
- Identifies significant research gaps in existing literature that this project aims to fill.

Chapter 3: Methodology

- Describes the datasets used: CoNLL-2003, CoNLL-2000, and Brown Corpus.
- Describes the preprocessing pipeline including subword alignment, tokenization, normalization, and masking strategies.
- Explains the proposed GAN-based architecture:
 - **Generator:** TinyBERT with a softmax head for token-level prediction.
 - **Discriminator:** TinyBERT with a binary classifier for token-level feedback.
- Describes the structure of the adversarial training loop, loss functions, and evaluation techniques.

Chapter 4: Results and Discussion

- Provides a comprehensive analysis of the model's performance on all three tasks.

- Presents key metrics such as accuracy, precision, recall, F1-score, and class-wise breakdowns.
- Discusses the implications of results and insights derived by reflecting on the patterns of errors.

Chapter 5: Conclusion and Future Scope

- Summarizes the contributions and results of the project.
- Suggests future directions, such as multilingual adaptation, deployment strategies, and advanced attention mechanisms.

CHAPTER 2

REVIEW OF LITERATURE

2.1. RELATED WORKS

Yu Tong et al. [1] described a unified GAN-based framework for sequence labeling, where a discriminator judges the accuracy of predicted tags rather than true or fake data. The authors' framework is composed of a BERT-based generator, a CRF decoder, and an attention-based discriminator, allowing for strong performance across tasks in a variety of languages.

Nicholas Popović and Michael Färber [2] presented EMBER; an efficient, and non-intrusive method for Named Entity Recognition in decoder-only language models (i.e. offspring of decoder-only transformers). Instead of fine-tuning, EMBER uses lightweight probing classifiers on hidden states and attention weights to classify both token-level entity types and spans in a streaming text generation scenario. The work achieved approximately 80–85% F1 scores, while having the potential to be up to 80× faster than conventional methods, making EMBER appropriate for on-the-fly processing and to complete semantic extraction in real-time.

Ayush Kumar et al. [3] proposed a sequence labelling method based on the BERT architecture for a dependency parser in Tamil, which is a morphologically rich language and has a free word-order. The method used was to tokenize each word with the BERT models, and then a machine learning algorithm is used to find dependency relations. For improving the knowledge about the input data that was labelled as a dependency parser, they leveraged techniques to address the imbalance of the data, oversampling methods such as SMOTE and ADASYN. The results were fed back into the Malt parser, implying ways that feature based methods and pretrained language models can support dependency parsing for low-resource languages.

Iker García-Ferrero et al. [4] conducted a comprehensive cross-lingual study of sequence labelling in zero-resource conditions. They compared two broad paradigms of transfer, one that is data-based (which is to say, machine translated and projecting annotated labels over), and the other being model based where they used multilingual language models in a zero-shot manner. Their findings demonstrated that high-capacity multilingual models outperformed the data based methods, as predicted, because the signal for the target language lost information in the translation process (as well as labelled multi-lingual data being gold copy). They also discussed how the data-based flow for sequence labelling was less impactful, overall, but a viable method for conducting sequence labelling in zero-resource contexts if large scale multi-lingual models were not available.

Tianwen Wei et al. [5] introduced Masked Conditional Random Fields (MCRF) for sequence labeling tasks to overcome the problem of invalid tag sequences created by the conditional random field (CRF)-based neural models. MCRF is able to constrain the valid tag sequences during training and decoding to enforce the tagging scheme constraints such as BIO. This new method is able to achieve good accuracy gains in the sequence labeling task to keep the added cost near-zero, and was able to successfully plug into the CRF-based system in which it was developed for.

Nuo Chen et al. [6] proposed a new pre-training approach as a solution to the objective gap between pre-training and fine-tuning in cross-lingual sequence-labeling tasks. They called the new pre-training task as Cross-lingual Language Informative Span Masking (CLISM) and also proposed a new contrastive learning approach called contrAstive-Consistency Regularization (CACR). The objective of CLISM is to align the pre-training with the sequence labeling objectives, while the CACR is intended to allow of more consistent representation across the parallel sequences. Their work shows much stronger performance than

previous approaches across multiple benchmarks and improved variations in few-shot performance, and therefore, greater cross-lingual transfer performance overall.

Marco Farina et al. [7] have tested a hallucination-free framework for sequence labeling for the efficient knowledge distillation from large encoder-decoder transformers. Their work responds to the growing need for more practical, compute-efficient distillation tactics, given the rapid growth of language models. The authors provide the evidence that they reach state-of-the-art results across sequence labeling benchmarks and validate the framework's performance in few-shot learning context, proving it both scalable and viable for practical implementation.

Luoxin Chen et al. [8] introduce Masked Adversarial Training (MAT) as a means of improving the robustness of sequence labeling models, by effectively amplifying the contextualization of features during adversarial training. Unlike standard adversarial training which adds word-level perturbations, MAT masks or replaces words and computes the adversarial loss from the modified input. They demonstrate significant improvements in accuracy and robustness on chunking and named entity recognition (NER) in many cases, showing that their model can competitively achieve state-of-the-art results using fewer parameters, and providing further gains by using a base model's contextual embeddings (e.g., ELMo).

Suyash Vardhan Mathur et al. [9] treated the SemEval-2024 Task 3 problem of Multimodal Emotion Cause Pair Extraction in dialogues as a sequence labeling problem with some encoders, BiLSTM for conversational context and a CRF layer for modeling utterance dependencies. Their proposed architecture approached 8th on the official leaderboard with F1 score of .1759 indicating sequence labeling

approaches could significantly improve multimodal emotion cause analysis.

Shuhe Wang et al. [10] proposed GNN-SL as a sequence labeling development framework for improved prediction that incorporates the nearest examples of a tag from the training set with Graph Neural Networks (GNN) to incorporate tagging examples, which they pulled from the training set. The GNN-SL framework links input words to their similar training set examples as a heterogeneous graph and informs the prediction process to provide appropriate input to long-tail, and minority classes. At the state of the art across benchmarks for Chinese Word Segmentation, Named Entity Recognition (NER) and Part-of-Speech tagging (POS).

Yekyung Kim [11] proposed a deep active learning framework for sequence labeling that couples uncertainty and diversity using gradient-based embeddings. The work extends the BADGE approach for image classification to sequence labeling tasks (such as Named Entity Recognition and slot-filling) by selecting diverse informative samples in gradient space and minimizing annotation effort while performing better than classical uncertainty and diversity sampling for most learned models across several datasets.

Zichun Yu et al. [12] introduced AutoSeq, a fully automatic prompting framework for sequence-to-sequence models. The framework eliminates the need for manually designed templates by replacing single label words with expressive label sequences of indefinite length. In the prompt-based fine-tuning by most researchers, labeling involves producing candidate sequences via the beam search algorithm and re-ranking them via contrastive probability (CP). Their extensive experiments showcase AutoSeq is a strong performer compared to both hand-crafted prompts and other automatic

Ziyang Huang et al. [13] proposed DiffusionSL, a new framework for sequence labeling based on a conditional discrete diffusion model. They model natural language sequences as conditional signals, and perform an iterative refinement of a noisy label sequence through a reverse diffusion process. To handle the nature of discrete tags, they developed the Bit-Tag Converter and Bit Diffusion Transformer. Even better, DiffusionSL shows better performance than autoregressive models and better than all current state-of-the-art baselines across multiple benchmarks by offering a more stable and progressive approach to sequence labeling.

Artem Shelmanov et al. [14] conducted a comprehensive review of the use of deep pre-trained models together with active learning (AL) for sequence tagging tasks. They examined Bayesian uncertainty estimation techniques, for example Monte Carlo dropout and BALD as part of the AL framework when using BERT and ELMo. They identified the best query strategies for different types of models, and showed how distilled versions of Transformers can substitute for their full-size versions in AL practices- benefiting computation efficiency without sacrificing model quality.

Xiaonan Li et al. [15] proposed SENTEE and TOKEE, two methods for enabling early-exits to accelerate BERT inference in sequence labeling tasks, like named-entity recognition (NER), part-of-speech tagging, and word segmentation. SENTEE performs sentence-level early exits based on uncertainty thresholds, while TOKEE performs token-level early exits using local context windows. A second fine-tuning stage with uncertainty-based halting was introduced to mitigate train-inference discrepancies. They demonstrated that their method can reduce inference cost by 66-75%, with little loss of task performance, while outperforming smaller compressed methods such as DistilBERT for equivalent speed-up conditions.

Henry Tsai et al. [16] proposed small-factored and efficient multilingual BERT-based sequence labeling models for part-of-speech tagging and morphological feature predictions in 48 languages. Their models are 6× smaller and 27× faster than standard BERT's multilingual model while still reporting state-of-the-art results, especially for low-resource languages or code-mixed text. The study validates the practical deployment of small multilingual models for real-world setting, particularly for CPU-constrained deployments, without affecting accuracy or performance.

Luoxin Chen et al. [17] put forward SeqVAT, an adaptation of Virtual Adversarial Training (VAT) for sequence-labeling tasks that incorporate a CRF layer. The implementation of VAT in traditional settings had challenges in sequence models as the label transition has dependencies. SeqVAT considers adversarial perturbations as part of learning while satisfying transition constraints during CRF decoding. The results for robustness and performance have improved for both supervised and semi-supervised options, and SeqVAT achieved performance above the baseline and existing VAT methods on tasks such as NER and chunking.

Karthik Raman et al. [18] have done a thorough analysis of how to best turn sequence tagging tasks into Seq2Seq tasks using large generative language models. They explored 14 different input-output transformation formats, across seven benchmarks, providing the Sentinel-based format that performed better than the widely used TaggedSpans format, with respect to accuracy and hallucination rates. Their empirical results with various scenarios in multilingual and zero-shot settings provide guidelines for selecting text-to-text conversion strategies when using Seq2Seq models for NER, SRL, and slot filling.

Tao Gui et al. [19] proposed Uncertainty-Aware Label Refinement (UANet) for sequence labeling tasks to overcome some of the limitations

of CRF-based methods in extracting longer-range label dependencies and decoding efficiency. The method first uses Bayesian neural networks to predict draft labels with uncertainty estimations, then refines just the uncertain labels through a two-stream self-attention methodology that considers both label and word-label dependencies. This fast, semi-automated, and parallelizable method allows it to process queries faster, together with superior accuracy on more than 8 public sequence labeling datasets, showing improvements over CRF and Seq2Seq-based techniques.

Peijie Jiang et al. [20] introduced a new model called BABERT, or Boundary-Aware BERT for Chinese sequence-labeling tasks like word segmentation, POS tagging, and NER. Without being dependent on registering lexicons manually to use the boundary information at hand, BABERT augments unsupervised statistical boundary features extracted from much larger corpora into the internal layers of BERT. Experimenting on 10 datasets, BABERT had improvements on all baselines (realistic improvements on competing Chinese PLMs on all tasks) further enhanced when combined with lexicons using a supervised boundary-based method.

Emile Chapuis et al. [21] investigated a hierarchical pre-training approach to sequence labeling in spoken dialog. Their study focuses on hierarchical transformer-based encoders, proposed the extension of the three standard pre-training objectives, including Masked Language Modeling (MLM), Generalized Autoregressive Pretraining (GAP), and incorporated discourse level features in multi-utterance scenarios. After evaluating on their newly created SILICONE benchmark for spoken language sequence labeling, they showed the potential to obtain competitive performance with fewer parameters compared to transformer-based models, and showed the benefits of pre-training specifically tailored to dialog level structure and data.

Fajri Koto et al. [22] proposed a top-down discourse parsing approach, reformulating RST parsing as a sequence labeling task. Their proposed scanning approach uses LSTM and pre-trained BERT to iteratively segment documents into discourse units, which simplified the process by eliminating decoders and reducing the search space for split points.

Zechuan Hu et al. [23] contributed a novel way to create a unified framework for zero-shot sequence labeling. They introduced minimum risk training to account for noisy pseudo-labeled data, coming from source models. As part of this, they created a new decomposable risk function to model relationships with imperfect predictions and true labels. The risk function was made trainable, which they could incorporate into a latent variable learning algorithm (using EM) and their model outperformed previous state of the art baselines across domains and languages on all 21 datasets for cross-lingual and cross-domain sequence labelling tasks.

Wei Liu et al [24] presented LEBERT, a framework, lexicon-enhanced BERT for Chinese sequence labeling task, that uses external lexicons directly in the BERT's intermediate show's transformer layers through a new Lexicon Adapter. This model contrasts to prior methods that fused lexicon as shallow sequence layer features as LEBERT permits a deeper, naturally nested, bottom level relationship with the lexicon knowledge and contextual embeddings. LEBERT achieved state-of-the-art results on ten datasets including Named Entity Recognition, Word Segmentation, and POS tagging to demonstrate the strength of deep lexicon integration.

Takuma Kato et al. [25] introduced a label component embedding framework for sequence labeling, specifically targeting fine-grained Named Entity Recognition. Instead of treating labels like B-Person or I-Location as atomic units, the method decomposes labels into shared

components (e.g., span indicators and entity types) and learns embeddings for each component. These embeddings are then combined to form full label representations. Their experiments on English and Japanese NER showed significant improvements, especially on rare or low-frequency labels, highlighting the value of modeling shared label structure.

Luoxin Chen et al. [26] proposed Masked Adversarial Training (MAT) to improve the robustness of sequence labeling models. Unlike standard adversarial training, MAT masks or replaces input words to encourage reliance on context rather than individual tokens. This enhances generalization and robustness against noisy inputs. MAT achieves strong results on tasks like chunking and NER, matching or surpassing state-of-the-art performance on CoNLL benchmarks with fewer parameters.

Tianwen Wei et al. [27] proposed MCRFs, a modification of the CRF-based sequence labeling models which tackled the constraint of producing illegal tag sequences in the BIO tagging model. MCRFs fix candidate transitions in training and decoding, thereby removing illegal sequences (like $O \rightarrow I$ - transition). To implement, there's very little cost and not much computational burden. Experimental results demonstrate, MCRFs not only solve the illegal path issue but also achieve the best results in general that outperform the CRF standard in different sequence labeling tasks, and achieve state-of-the-art results on multitude of Chinese NER datasets.

Yuqian Zhao et al. [28] put forward the MLSU framework for unifying NER into flat, nested, and discontinuous NER via a multi-level semantic approach, which uses a novel framework of tracking word relationships claimed as their multi-level semantics. MLSU takes an encoder-decoder BART-based architecture, which adds sequence-level fuse embedding (which is a combination of GloVe embedding and label

aware attention embedding), and a grid-level semantic layer consisting of conditional layer normalization (CLN) and Dilated convolutional neural networks (CNNs). Using these embeddings and their grid-level semantic models, MLSU can model the complex relationships of word-word and word-label to allow for more human-like recognizing of entities in complex structures. Experimental results over six benchmarks demonstrate that MLSU always ties or outperforms state-of-the-art experiments.

Pingchuan Ma et al. [29] proposed XBiLSTM-CRF, a Named Entity Recognition (NER) model designed for the cybersecurity sector, which is built on a combination of Bidirectional LSTM and Conditional Random Fields (CRF). Unlike conventional BiLSTM-CRF models, the xBiLSTM-CRF is constructed by concatenating embeddings into BiLSTM output before it is sent to the CRF layer and allows the model to harness its ability to recognize vocabulary particular to the domain. The model was tested on an open-source cybersecurity dataset that was manually relabeled, and modeled six types of entities (e.g., software, attack, modifier). The xBiLSTM-CRF whole achieved the best precision, recall, and F1 scores overall based upon the chosen baselines and provides evidence of its effectiveness to extract structured knowledge from unstructured cybersecurity related text.

Naji Esmaail et al. [30] executed a systematic literature review (SLR) of Named Entity Recognition (NER) in user generated text on social media platforms, namely X (formerly Twitter). The review highlights 67 selected papers published from 2015 to 2023, in particular WNUT benchmark datasets. The paper identifies a dozen different types of approaches (ML, DL, NLP, hybrid) and research evaluation metrics and highlights challenges like label imbalance, noisy and sparse data, newly emerging entities, and contextual constraints. It further proposes future research directions such as improved performance in few-shot learning, accept more multimodal inputs,

robustness in noisy environments, and that NER models can and must improve.

Mingye Yang et al. [31] developed BBC-Ap, a domain-specific NER model for the aviation products domain, integrating BERT, BiLSTM, and CRF layers to extract entities from complex, technical documents. Addressing challenges like specialized terminology and lack of datasets, the authors created an ontology-based annotation schema and a custom ApNER dataset with over 8,000 entities. The model achieved 92.1% F1-score on the domain dataset and performed competitively on standard benchmarks (e.g., CoNLL-2003, CoNLLpp). Additionally, the extracted entities and relationships were used to build a Neo4j-based knowledge graph, supporting intelligent data utilization in aviation manufacturing.

Gang Yang et al. [32] proposed a Residual BiLSTM-CRF model to improve performance in Named Entity Recognition (NER) tasks across English and Chinese datasets. Inspired by ResNets, the model introduces novel residual BiLSTM blocks with localized skip connections and auxiliary layers (e.g., layer normalization, ReLU, dropout) to enhance long-term dependency learning without external knowledge. Evaluated on standard benchmarks (CoNLL-2003, MSRA, Weibo, OntoNotes), the model achieves state-of-the-art F1 scores, especially when paired with dynamic BERT embeddings. Extensive ablation and structural analyses confirm the efficiency and robustness of the proposed architecture compared to conventional and existing residual BiLSTM variants.

Huiming Zhu et al. [33] proposed FSeqC, a sequence-to-sequence framework for fine-grained Named Entity Recognition (NER), designed to handle over 100 entity types efficiently. To overcome the limitations of manual annotations and coarse-grained tagging, the authors employ distant supervision by aligning Wikipedia hyperlinks with Freebase

types, and introduce a BIO-based sec-part tagging scheme combining position and hierarchical class attributes. The model utilizes a Bi-LSTM encoder, self-attention mechanism, LSTM decoder, and a CRF layer to enhance contextual understanding and enforce logical tag transitions. Experiments on large-scale datasets show that FSeqC significantly outperforms state-of-the-art models in precision, recall, and F1, especially when enhanced with cleaner annotated data.

Chang Tian et al. [34] proposed SMCS, a novel pipeline model for few-shot Named Entity Recognition (NER) that addresses two major issues: repetitive training of span detectors and sample dependency in entity classification. SMCS introduces a pre-trained “Steppingstone” span detector based on Wikipedia data to eliminate the need for redundant training on basic features. For entity classification, SMCS leverages machine common sense by using LLM-generated entity-type definitions instead of relying on few-shot samples to define class centroids. This approach reduces overfitting and improves generalization. Experiments on Few-NERD and Cross-Dataset benchmarks show that SMCS outperforms or matches strong baselines (including ChatGPT) in fine-grained and cross-domain few-shot settings, offering an efficient and effective alternative without large model overhead.

Qi Cheng et al. [35] introduced a novel prompting method tailored for few-shot Named Entity Recognition (NER) using Large Language Models (LLMs). Unlike generic prompts designed for generation tasks, their method structures prompts into three components: task definition, few-shot demonstrations, and output format constraints. They also incorporate a feedback-based strategy that selects improved few-shot examples based on LLM output errors, refining prompt quality. This approach significantly enhances few-shot NER performance, demonstrating LLMs’ capability to adapt to structured prediction tasks with minimal supervision.

Xiu-Qin Pan et al. [36] proposed LGENER, a novel method for Named Entity Recognition (NER) in ancient ethnic Chinese texts, integrating lattice structures, BERT, and Generative Adversarial Networks (GANs). LGENER enhances the FLAT model by incorporating relative positional encoding and an auxiliary entity boundary detection task. A CNN-based discriminator evaluates label sequences generated by a BERT-Transformer-XL-CRF generator to optimize predictions via adversarial learning. Experiments on Yi and Bai ethnic datasets and general Chinese corpora (e.g., Resume, Weibo) show that LGENER significantly improves F1 scores, especially for complex and long-distance dependencies. The model achieves an F1 of 86.86% on Yi texts and demonstrates strong generalizability across domains.

Shuai Guo et al. [37] introduced DSW (Double Sliding Window), an improved content-defined chunking algorithm for data deduplication in ocean observation systems. DSW enhances traditional block-level deduplication by applying two fixed-size sliding windows and a dual-condition hash-based segmentation mechanism to ensure block sizes remain within a controlled range. The method improves deduplication efficiency while minimizing memory overhead by producing more uniformly sized data chunks. A Markov model is also integrated to predict chunking patterns, further optimizing performance. Experimental results on real-world marine datasets show that DSW outperforms existing methods like BSW and TTTD in re-deletion ratio, block count consistency, and resource efficiency.

Nai Ding et al. [38] introduced a framework for sequence chunking through neural encoding of ordinal positions, proposing that the brain organizes sensory events into chunks while tracking each event's position within a chunk. This improves conventional models by showing that low-frequency neural activity signals both chunk boundaries and positions. The framework integrates EEG, MEG, and ECoG findings,

revealing both domain-specific and domain-general networks in chunking. It further suggests that chunk-tracking activity causally supports predictive chunking by maintaining positional codes. Experiments across language, motor, and sequential tasks confirm that neural activity reliably tracks hierarchical chunk structures.

Philipp Musfeld et al. [39] studied how working memory grouping influences chunk formation in long-term memory using the Hebb repetition paradigm. They proposed that chunk formation relies on recognizing repeated patterns within sequences, not their fixed positions. Across six experiments, they enhanced prior work by manipulating sequence salience through spatial grouping and color coding. Results showed participants learned repeated sequences flexibly at any position when the structure was recognizable, challenging position-dependent learning models. Increased salience improved recognition and chunk formation, especially for sequences appearing in varying positions, demonstrating that working memory grouping shapes long-term memory chunking and refining repetition learning models.

Xiaoqi Jiao et al. [40] proposed TinyBERT, a lightweight Transformer model distilled from BERT, optimized for deployment on resource-limited devices. They introduced a two-stage knowledge distillation framework—first during pretraining, then during task-specific fine-tuning—to transfer both general and task-level knowledge from a large BERT teacher model. This approach improves upon prior work by aligning both the embeddings and intermediate representations across all Transformer layers. Experimental results showed that the 4-layer TinyBERT retained over 96.8% of BERTBASE’s GLUE performance while being $7.5\times$ smaller and $9.4\times$ faster. A 6-layer TinyBERT matched BERTBASE’s performance, confirming TinyBERT’s effectiveness in balancing compactness and accuracy.

Kyungtae Lim and Jungyeul Park [41] introduced a multiview learning approach for POS tagging that combines multihead-attention character-level representations at both the word and sentence level. By jointly training these through a meta-BiLSTM, their model captures both local and global subword patterns, addressing weaknesses in earlier concatenation-based methods. Tested on the CoNLL 2018 dataset, their system delivered state-of-the-art results in nine languages, with especially strong improvements for morphologically rich languages like Ancient Greek, where it gained +8.89 accuracy points over prior models.

Alim Murat and Samat Ali [42] introduced a POS tagging model for low-resource, morphologically rich languages like Uyghur, Uzbek, and Kyrgyz. Their approach combines affix-based word embeddings (using character n-grams) with multi-head attention on top of a BiLSTM-CRF, capturing both subword patterns and contextual dependencies. Tested on a custom MultiPOS_ukg dataset and Turkish Treebank, it outperformed baselines by up to 5.36%, showing strong handling of OOV words and ambiguous forms — proving affix-aware features and attention mechanisms can significantly improve POS tagging for under-resourced, agglutinative languages.

Latofat Bobojonova et al. [43] introduce BBPOS, the first BERT-based part-of-speech (POS) tagging models for Uzbek, addressing the scarcity of NLP resources for this low-resource, morphologically rich language. They develop a new UPOS-tagged dataset comprising 500 sentences in both Latin and Cyrillic scripts. Fine-tuning two monolingual Uzbek BERT models on this dataset, BBPOS achieves an average accuracy of 91%, surpassing both multilingual BERT and existing rule-based taggers. Notably, the models effectively capture affix-induced POS changes and demonstrate context sensitivity, marking a significant advancement in Uzbek NLP.

Juan Manuel Coria et al. [44] present a study on how well multilingual BERT handles continual learning for sequence labeling tasks like slot-filling and named entity recognition. Using nine languages from the MultiATIS++ and MultiCoNER datasets, they train the model sequentially and find that although some forgetting happens, the model retains a good amount of cross-lingual knowledge. Even better, a small amount of retraining can recover most of the lost performance. The paper also points out that current evaluation methods may not fully capture the nuances of continual learning, suggesting the need for better metrics.

Baojun Wang et al. [45] introduce DyLex, a plug-in framework designed to seamlessly integrate large-scale, frequently updated lexicons into BERT-based sequence labeling tasks such as Named Entity Recognition (NER), Chinese Word Segmentation (CWS), and slot-filling. Unlike traditional methods that rely on static word embeddings, DyLex employs word-agnostic tag embeddings, eliminating the need for retraining when lexicons are updated. It also incorporates a supervised denoising mechanism to filter out noisy matches and utilizes a column-wise attention mechanism to effectively fuse lexical features with BERT representations. Evaluations across ten datasets demonstrate that DyLex achieves state-of-the-art performance, highlighting its robustness and adaptability in handling dynamic lexicons.

Ben Athiwaratkun et al. [46] introduce a generative framework that unifies sequence labeling and sentence-level classification by leveraging a shared natural language output space. This approach naturally incorporates label semantics and facilitates knowledge sharing across tasks. The model demonstrates strong performance across various settings, achieving state-of-the-art results in few-shot slot labeling—improving 5-shot accuracy from 75.0% to 90.9% and 1-shot from 70.4% to 81.0%. It also shows significant gains in low-resource scenarios and

maintains competitive results on high-resource tasks, setting a new benchmark on the SNIPS dataset.

Wei Liu et al. [47] present LEBERT, a novel framework that integrates external lexicon knowledge directly into BERT's architecture for Chinese sequence labeling tasks. By introducing a Lexicon Adapter layer between BERT's Transformer layers, LEBERT enables deep fusion of lexicon features, enhancing the model's understanding of Chinese linguistic structures. Evaluated across ten datasets encompassing Named Entity Recognition, Word Segmentation, and Part-of-Speech Tagging, LEBERT achieves state-of-the-art performance, demonstrating the effectiveness of embedding lexicon information within BERT's lower layers.

Kamil Bujel et al. [48] explore how transformer-based sentence classifiers can be adapted for zero-shot sequence labeling without any token-level supervision. They identify that traditional zero-shot methods underperform with transformers due to the dispersion of information across multiple attention layers. To address this, the authors introduce a modified soft attention mechanism that sharpens attention weights, enabling the model to better focus on relevant tokens. Their approach significantly outperforms existing methods like LIME, achieving new state-of-the-art results in zero-shot sequence labeling while maintaining strong sentence-level classification performance. This work highlights the potential of repurposing sentence-level transformers for token-level tasks without additional supervision.

Kun Luo et al. [49] introduce Landmark Embedding, a novel chunking-free method for retrieval-augmented long-context language models. Unlike traditional approaches that divide text into chunks—potentially disrupting semantic coherence—this method maintains the integrity of the entire context, enabling the generation of high-quality

embeddings for fine-grained information spans. A position-aware objective function is employed to emphasize critical information boundaries, and a multi-stage learning algorithm leverages both real and synthetic data for efficient training. Experimental results demonstrate that Landmark Embedding significantly enhances the performance of models like LLaMA-2 and ChatGPT across various long-context tasks, outperforming existing retrieval methods.

Hongjin Qian et al. [50] propose Chunking-Free In-Context (CFIC) retrieval for Retrieval-Augmented Generation (RAG) systems, eliminating the need to split documents into chunks. Unlike traditional methods that risk losing semantic coherence and adding irrelevant info, CFIC uses encoded hidden states of whole documents for precise evidence retrieval. It employs two decoding strategies—Constrained Sentence Prefix Decoding and Skip Decoding—to improve efficiency and maintain evidence quality. Evaluations on open-domain QA datasets show CFIC outperforms chunking-based methods in accuracy and efficiency, providing a more streamlined approach for grounding RAG responses.

Li et al. [51] developed a new data augmentation approach for sequence-labeling tasks, called Task-Aware Generation (TAG). In TAG, the authors use a generative model to create synthetic labeled sequences that are semantically coherent and relevant to the task. When combined with existing seq labeling models, the authors showed improvements in performance across all their benchmarks, with their approach being particularly impactful in low-resource settings. Their findings suggest that task-aware data augmentation is an effective way to improve model generalization and robustness.

Fu et al. [52] took on the challenge of domain shift in Generative Adversarial Networks (GANs) by developing an unsupervised learning framework that allows for generalization across domains. They trained

GANs with attention to the learning of domain-invariant features to generalize better when deploying to unobserved domains. The experiment demonstrates the model's ability to perform generation and classification tasks of relatively high quality when faced with considerable domain shift, demonstrating suitability for strong unsupervised learning.

Han et al. [53] investigated the use of pretrained language models along with adversarial training to increase robustness in Chinese Named Entity Recognition (NER) systems. The idea is that the model will become robust to input perturbations and noise by adding adversarial examples during training time. They showed that this combination significantly enhanced NER performance - especially for conditions of limited annotated samples - indicating that robustness is critical in real world NLP applications.

Shen et al. [54] explored more broadly the impact of adversarial training on the cross-domain generalization capabilities of an NLP model. Including adversarial examples in the training process nudges models to learn a more generalized representation which is less sensitive to domain specific features. Findings suggest that using these training strategies led models to improve significantly on an out-of-domain dataset, indicating a potential path toward more flexible and robust NLP models.

Liu et al. [55] presented a method for examining reading comprehension systems' robustness through adversarial examples. The authors created an intentionally misleading input using changes to the text that ultimately do not affect the correct answer, thereby getting an insight into how well these systems understand language. The authors found that many of the state-of-the-art models are vulnerable to these adversarial inputs and highlight the importance of creating a more robust reading comprehension system.

Wolf et al. [56] introduced the Transformers library, an open-source platform that offers a large selection of pretrained transformers built for natural language processing tasks. The creators of the library wanted to democratize NLP by making user-friendly interfaces, as well as thorough documentation, to promote easy use for research and industry. Its widespread use is a prime example of providing accessible tools to develop the field of NLP.

2.2. RESEARCH GAPS

- Over-reliance on extra-large models (e.g., BERT) for adversarial training without modification for lightweight deployments.
- Softmax decoders are rarely used in adversarial framework.
- Lack of a generic multi-task frameworks to encapsulate NER, chunking, and POS tagging.
- Very few studies on adversarial training for tasks other than NER.
- Limited use of lightweight transformers (e.g., TinyBERT) in sequence labeling with GANs.
- No benchmark model that uses only standard English datasets in an adversarial way.
- The CRF complexities and training overheads continue to be a major bottleneck.
- There are many large-scale trade-offs between model compactness and multi-task performance that are under-researched.
- Insufficient Exploration of Adversarial Robustness Across Tasks.

2.3. SUMMARY

This chapter surveyed the literature on sequence labeling, particularly on models such as BERT-based classifiers, BiLSTM-CRFs, and SLGAN (Sequence Labeling GAN). In this review, we provided an overview of the performance, strengths, and weaknesses of each model type that has been examined about sequence labeling tasks. Special emphasis was placed on applications for NER and Chunking. The review reveals a research gap in models that offer both high accuracy and computational efficiency. It further discusses how TinyBERT and GANs have been used

in other NLP tasks, helping to contextualize the novelty of your approach. This survey justifies your design choices and motivates the shift toward lightweight, adversarial learning-based models.

CHAPTER 3

METHODOLOGY

3.1. INTRODUCTION

This chapter outlines the architectural framework, training procedures, and technological setup used to develop a lightweight yet robust adversarial model for sequence labeling tasks in NLP, including Named Entity Recognition (NER), Part-of-Speech (PoS) Tagging, and Syntactic Chunking. The approach integrates a **Generative Adversarial Network (GAN)** paradigm with **TinyBERT**, a compressed version of the BERT model, to create a multi-task model capable of performing token-level classification with high efficiency.

To meet the requirements of modern NLP applications where both accuracy and computational efficiency are critical, this methodology leverages three core components:

- **Dataset selection and preprocessing**, using benchmark English-language datasets that provide standardized annotations for sequence labeling tasks.
- **Model design**, where a TinyBERT-based generator and discriminator interact adversarially to improve labeling robustness.
- **Training and evaluation protocols**, employing adversarial loss functions alongside supervised learning techniques for fine-grained performance across multiple tasks.

This chapter first introduces the conceptual underpinnings of GANs and the role of TinyBERT in building compact, contextual models. It then details the preprocessing steps, architectural modules, training procedures, evaluation strategies, and tools used to implement the model.

3.2. Generative Adversarial Networks (GANs)

Generative Adversarial Networks (GANs) are a class of machine learning frameworks where two models—the generator and the discriminator—

compete in a minimax game. Traditionally used in image generation, GANs have recently shown promise in sequence tasks by enhancing generalization and robustness.

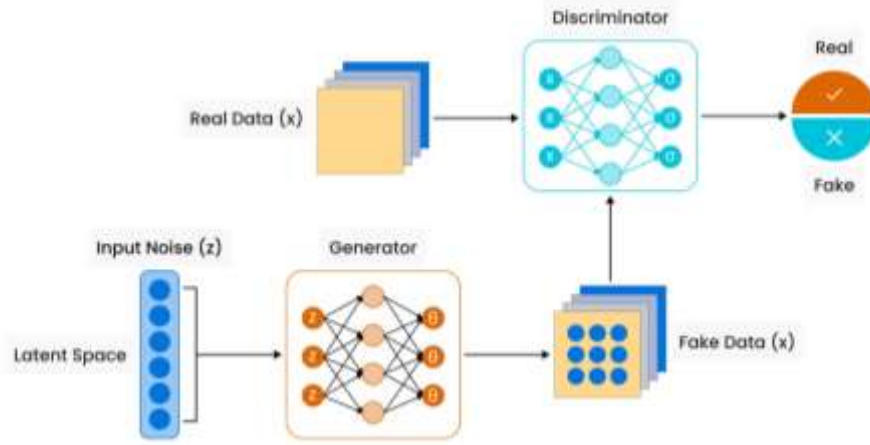


Figure 3.2 Generative Adversarial Network

- **Generator:** Attempts to produce outputs (in this case, token labels) that are as realistic as possible, mimicking the distribution of true labels.
- **Discriminator:** Acts as a binary classifier that distinguishes between real (ground-truth) and generated (predicted) label sequences at the token level.

In the context of sequence labelling, the GAN paradigm provides a feedback loop where the generator continuously learns from the discriminator's judgments, making it capable of learning subtle distinctions in token-level patterns. This adversarial setup helps:

- Combat overfitting on small datasets.
- Improve the model's ability to generalize to rare or ambiguous tokens.
- Enforce consistency between label distributions and contextual information.

Unlike traditional CRF-based approaches, this model employs a softmax classifier at the output layer, simplifying the decoding process while maintaining competitive accuracy.

3.3. TinyBERT

TinyBERT is a distilled version of BERT (Bidirectional Encoder Representations from Transformers) designed to retain most of BERT’s representational power while reducing model size, training time, and inference latency.

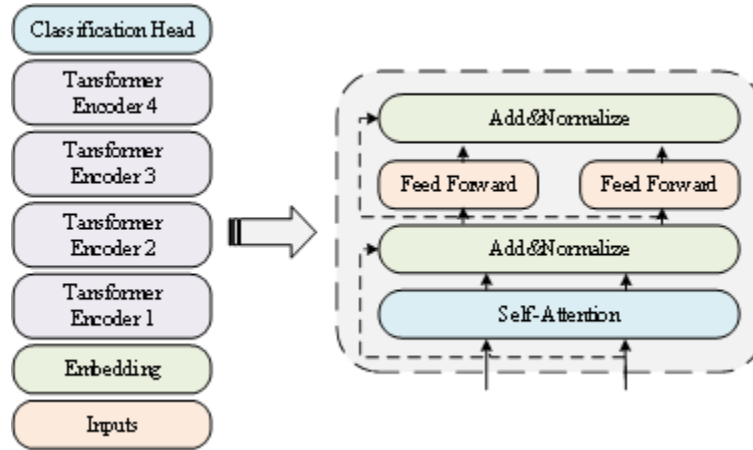


Figure 3.3 TinyBERT Architecture

Key properties of TinyBERT:

- **6-layer encoder**, as opposed to BERT-base’s 12 layers.
- **Fewer hidden dimensions** (e.g., 312 vs. 768).
- **Distilled knowledge** from a larger teacher model using task-specific and layer-wise distillation techniques.
- **Significant memory and speed gains** without major performance loss on NLP benchmarks.

In this methodology, TinyBERT is used as the encoder backbone for both the generator and the discriminator, chosen for:

- Efficiency in resource-constrained environments (e.g., real-time systems, mobile devices).
- Adequate contextual understanding for token classification.
- Compatibility with the HuggingFace Transformers ecosystem.

TinyBERT’s ability to model local and global dependencies across tokens makes it well-suited for sequence labeling tasks, especially when paired with appropriate tokenization and subword alignment strategies.

3.4. DATASETS

To ensure the proposed sequence labeling model performs effectively across core NLP tasks, a carefully selected set of English-language benchmark datasets was used. Each dataset targets a specific labeling task and follows standardized annotation formats that support token-level classification. The following datasets were integrated into the training and evaluation pipeline:

1. **CoNLL-2003:** A widely used dataset for Named Entity Recognition (NER), containing annotated newswire text with entity types such as persons, organizations, locations, and miscellaneous entities. Gathered from Newswire articles from Reuters RCV1.
2. **CoNLL-2000:** Focused on Syntactic Chunking, this dataset includes phrase-level annotations (e.g., noun phrases, verb phrases) extracted from the Penn Treebank, making it ideal for shallow parsing tasks. Created from Sections of the Wall Street Journal corpus from the Penn Treebank.
3. **Brown Corpus:** One of the earliest and most comprehensive corpora for Part-of-Speech (PoS) Tagging, covering a diverse range of topics and genres. It provides token-level PoS annotations useful for grammatical analysis and linguistic variability. Created from A diverse set of texts across genres like fiction, government, and news.

These datasets were preprocessed and aligned with subword tokenization strategies to suit the TinyBERT model. The training framework was designed for token-level prediction, ensuring compatibility across tasks. By leveraging these well-established resources, the system is capable of generalizing effectively within English-language domains, supporting robust evaluation across NER, Chunking, and PoS tagging benchmarks.

3.5. PROPOSED METHODOLOGY

The development of this GAN-based sequence labeling framework involves a structured and iterative process designed to ensure accuracy, efficiency, and robust generalization. The methodology is divided into the following stages, each contributing to the development of a high-performing token-level labeling system for NLP tasks such as NER, PoS tagging, and Chunking.

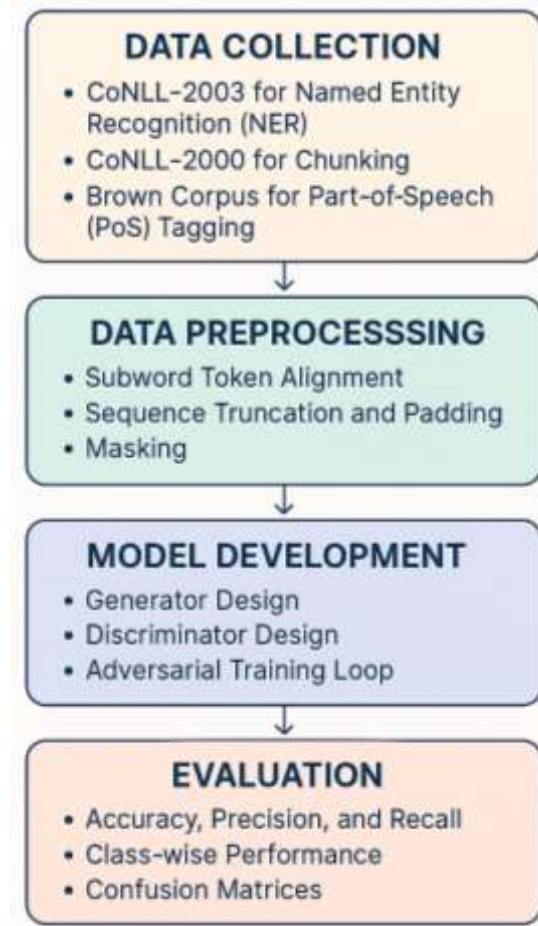


Figure 3.5 Proposed Methodology

1. Data Collection:

- Annotated datasets were selected from well-established English NLP benchmarks. These include:
 - **CoNLL-2003 for Named Entity Recognition (NER)**: Provides BIO-tagged newswire text covering entities such as persons, organizations, locations, and miscellaneous.

- **CoNLL-2000 for Chunking:** Offers syntactic chunk labels such as NP (noun phrase), VP (verb phrase), and PP (prepositional phrase).
- **Brown Corpus for Part-of-Speech (PoS) Tagging:** Includes token-level grammatical tags across multiple genres.
- These datasets collectively support the model's ability to learn syntactic and semantic token-level structures effectively across various tasks.

2. Data Preprocessing:

• Subword Token Alignment

- Since TinyBERT uses subword tokenization, each label must be carefully aligned with the corresponding token(s). Only the first subword of each word is labeled; others are masked during training.

• Sequence Truncation and Padding

- All input sequences are padded or truncated to a fixed maximum length (e.g., 128 tokens). Special tokens like [CLS] and [SEP] are handled using HuggingFace's tokenizer.

• Masking

- To prevent errors from padded tokens or misaligned subwords, masking is applied. Tokens without valid labels (including padding) are assigned a loss mask value of -100.

• Tokenization

The datasets are tokenized using TinyBERT's pretrained tokenizer with support for:

- Word-piece token mapping via `word_ids()`
- Maintenance of original word boundaries for accurate label projection

• Normalization

Text normalization ensures consistency across tokens before embedding:

- Lowercasing all tokens to reduce case sensitivity

- Removing unnecessary punctuation, except where grammatically or semantically meaningful
- Handling contractions or irregular terms, especially in informal or news-like text

3. Model Development:

• Generator Design

The generator uses a TinyBERT encoder followed by a softmax classification head:

- Outputs a label for each token
- Trained using cross-entropy loss, with special handling for padding and subwords
- Provides initial label predictions for input sequences

• Discriminator Design

The discriminator is also based on TinyBERT and acts as a token-level binary classifier:

- Input: Text sequence + label sequence (real or generated)
- Output: Binary score per token (correct vs. incorrect label)
- Trained using binary cross-entropy loss

• Adversarial Training Loop

The model is trained using an alternating adversarial procedure:

- The generator predicts label sequences
- The discriminator evaluates whether each label is likely correct
- The generator is updated using both supervised and adversarial signals
- Both components are trained iteratively to improve labeling accuracy and robustness

4. Evaluation:

Model performance is evaluated using:

- Accuracy, precision, recall, and F1-score
- Class-wise performance analysis (e.g., I-ORG, I-MISC, B-PER)
- Emphasis on rare and ambiguous tokens to ensure generalization

3.6. ARCHITECTURE DESIGN

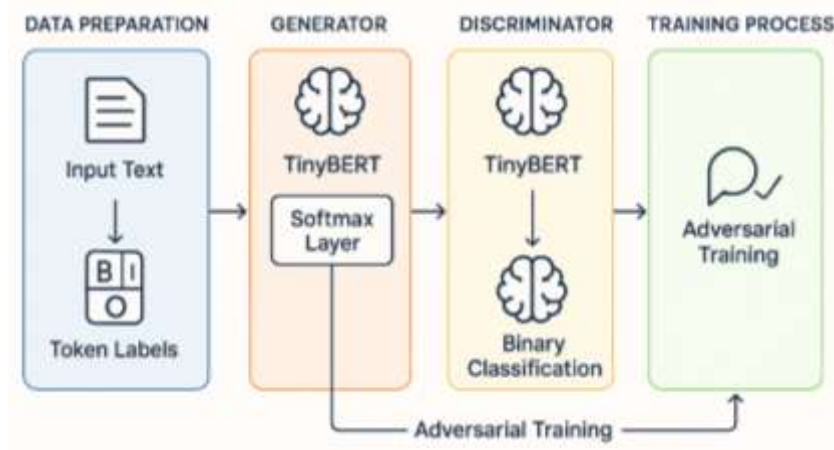


Figure 3.6 Architecture Design

The proposed architecture is a GAN-based sequence labeling system designed for English-language NLP tasks such as Named Entity Recognition (NER), Part-of-Speech (PoS) Tagging, and Syntactic Chunking. The system integrates a TinyBERT-based generator and discriminator, working in tandem through an adversarial training loop. The model is optimized for token-level prediction tasks and is structured across four main modules:

1. Data Preparation

In the first step, the raw annotated text data are transformed into a machine-readable format. Both the generator and discriminator must recognize and follow the same format. Processing steps include the following:

- **Input:** Raw sentences and their corresponding BIO-encoded token labels, extracted from:
 - CoNLL-2003 (for NER)
 - CoNLL-2000 (for Chunking)
 - Brown Corpus (for PoS Tagging)
- **Preprocessing Steps:**
 - **Subword Tokenization:** Each input sentence is tokenized using TinyBERT’s tokenizer, which may split words into

multiple subword tokens. For labeling purposes, the model only considers the first subword of each word.

- **Label Alignment:** Labels are aligned with their corresponding tokens using `word_ids()` in order to correctly map each label.
- **Masking:** The model is masking the non-labeled tokens (e.g. padded positions and subwords) based on a pre-defined special index (e.g. -100) so not included in the loss.
- **Padding and Truncation:** Input sequences are padded or truncated to a fixed maximum length (e.g., 128 tokens) to ensure consistent batch processing.
- **Normalization:** The normalization step is mainly for the token with respect to lower case, removal of punctuation, and spacing so that formatting is consistent across the input data.

This step is designed to help ensure the model is being provided with clean and consistent input for further training and inference.

2. Generator Module

Being a token-based classifier, the generator module can be thought of as a token-level "classifier" which predicts labels for each input.

- **Core Engine:** A TinyBERT encoder since it has a relatively small size and is very efficient for learning contextual dependencies.
- **Output Layer:** A softmax-based classification head to yield a probability distribution for every possible label for every token, identifying the token label probabilities.
- **Learning Objectives:**
 - **Cross-Entropy Loss:** The model is trained in a supervised manner to minimize the difference between predicted labels and ground-truth annotations.
 - During training, the model uses Cross-Entropy Loss for training in which only the unmasked (valid) tokens are used to generate the loss for the generator.

The generator is essentially a sequence labeler learning how to correctly label sequences based on the context around it.

3. Discriminator Module

The discriminator acts as a token-level verifier that evaluates the quality of labels assigned by the generator.

- **Core Engine:** Another TinyBERT encoder, identical in architecture to the generator but trained with a different goal.
- **Input:** The original input sentence combined with a label sequence—either from ground truth or predicted by the generator.
- **Output Layer:** A binary classifier head that outputs, for each token, a score indicating whether the label is real (correct) or fake (generated).
- **Learning Objective:**
 - Binary Cross-Entropy Loss: The discriminator is trained to maximize its ability to distinguish between true and generated labels at the token level.

This module provides fine-grained feedback to the generator, identifying precisely where label predictions are incorrect.

4. Adversarial Training Process

This is the core of the GAN setup, where both models improve iteratively.

- **Training Loop:**
 - The generator receives input text and outputs a sequence of predicted token labels.
 - The discriminator evaluates the real and predicted label sequences and provides token-level feedback.
 - The discriminator is trained to improve its classification performance.
 - The generator is updated with two losses
 - Supervised loss (cross-entropy with ground-truth labels)
 - Adversarial loss (to fool the discriminator)

- Goal: Enable the generator to produce increasingly accurate label sequences that are indistinguishable from ground-truth annotations, leading to improved performance even on rare or ambiguous inputs.g Process

3.7. TOOLS & TECHNOLOGIES

The implementation of the proposed GAN-based sequence labeling model utilizes a combination of programming tools, deep learning libraries, and utilities for data handling, training, and evaluation. The technologies were selected for their compatibility, performance, and support for token-level NLP tasks.

- **Programming Language**

- a) **Python:** The core language used for data preprocessing, model development, training loops, and evaluation. Python's ecosystem provides extensive support for NLP and deep learning through specialized libraries.

- **Libraries and Frameworks**

- a) **HuggingFace Transformers:** Utilized for loading and fine-tuning the pretrained TinyBERT model. It provides seamless access to tokenizers, pretrained weights, and sequence labeling heads.
- b) **Datasets:** Utilized for loading and fine-tuning the pretrained TinyBERT model. It provides seamless access to tokenizers, pretrained weights, and sequence labeling heads.
- c) **PyTorch:** The primary deep learning framework used for building the generator and discriminator models, managing training steps, loss functions, and backpropagation.
- d) **Scikit-learn:** Used for evaluation metrics including precision, recall, F1-score, and confusion matrix generation.
- e) **Matplotlib & Seaborn:** Employed for visualizing performance trends, metric comparisons, and confusion matrices.

- **Platform and Environment**

a) Google Colaboratory (Colab): The development and training of models were conducted in Google Colab, which provides access to free GPU/TPU resources, making it suitable for iterative experimentation with large models. It also allows integration with Google Drive for dataset storage.

- **Data Preprocessing Tools**

a) NLTK and spaCy: Used for auxiliary preprocessing tasks such as token-level cleaning, normalization, and sentence segmentation to prepare the raw datasets before tokenization.

b) HuggingFace Tokenizer (TinyBERT): Ensured precise word-piece alignment and label projection, essential for handling subword-based token classification.

- **Environment Management**

a) python-dotenv: Utilized to securely manage environment variables such as paths, hyperparameters, and model checkpoints during development.

- **Storage and Logging**

a) Google Drive: Used for storing intermediate checkpoints, training logs, and result files during and after model training.

3.8. SUMMARY

This chapter describes the design and development of the proposed model, which uses a TinyBERT-based Generator with a softmax classification head, and a TinyBERT-based Discriminator that evaluates predicted token labels. The model is trained adversarially using a combination of supervised (cross-entropy) and adversarial (binary classification) losses to improve robustness. It details the preprocessing steps such as token alignment, subword masking, padding, normalization, and the use of `word_ids()` for label alignment. The datasets used—CoNLL-2003 (NER), CoNLL-2000 (Chunking), and the Brown Corpus (PoS tagging)—are introduced, with justification for their use in evaluating performance across different labeling tasks.

CHAPTER 4

RESULTS AND DISCUSSION

4.1. PERFORMANCE METRICS

The performance of the proposed GAN-based sequence labeling system was evaluated across three core NLP tasks: Named Entity Recognition (NER), Syntactic Chunking, and Part-of-Speech (PoS) Tagging. The evaluation was conducted using accuracy, precision, recall, and F1-score, with both macro and weighted averages to account for class imbalance.

Table 4.1 Performance Metrics of the Proposed Methodology

Task	Accuracy	Precision	Recall	F1-Score
NER	95.41%	95.54%	95.41%	95.43%
Chunking	91.12%	89.53%	91.12%	90.18%
PoS Tagging	89.24%	88.99%	89.69%	89.24%

These metrics provide a comprehensive view of the model’s performance across tasks. High weighted F1-scores in NER and Chunking indicate good generalization. Strong performance in PoS Tagging indicates reliable syntactic labelling, especially on Brown Corpus.

4.2. DISCUSSION

The model proposed that uses TinyBERT in both the generator and in the discriminator of a GAN-based approach, has shown good token-level classification results, without having to leverage large-scale models like BERT or cumbersome decoders like CRF.

- **NER TASK:**

- The model achieved over 95% weighted F1-score on the CoNLL-2003 dataset.
- Excellent recognition of common entities like PER, ORG, and LOC.

- Minor weaknesses were noted in less frequent or more ambiguous labels such as MISC.
- **Chunking Task:**
 - Accuracy and F1-scores surpassed 90%, with high performance on core phrase types such as B-NP, B-VP, and I-NP.
 - Low recall was observed on rare chunk types like I-SBAR and I-ADVP, due to class imbalance.
- **PoS Tagging Task:**
 - Despite Brown Corpus' diversity, the system achieved ~89% accuracy and robust macro-level performance.
 - Tags like NN, VB, DT, and IN were classified with very high precision and recall.
 - Predictive challenges remained for underrepresented tags like UH, FW, and SYM, which received near-zero F1 in some cases.

These results reflect the benefits of adversarial training in improving generalization and correcting boundary-level misclassifications. The discriminator, by offering token-level feedback, pushed the generator to refine its predictions even for less frequent or ambiguous cases.

Overall, the system exhibits a high level of reliability and flexibility of tasks with computational performance based on the use of TinyBERT and softmax-based decoding. In addition, this study provides compelling evidence that GAN-based learning is feasible for real-world applications in sequence labeling tasks especially when few resources or few labeled data are available.

4.3. SUMMARY

This chapter presents the ability of the model for validation and test sets using a variety of performance metrics: accuracy, precision, recall, F1-score, and support per every label. With respect to visual results, the bar plots and confusion matrices substantiate the model achieving a very good classification for tags with frequency and managing complex boundaries with labels. The comparisons also show your model

obtaining results like, or better than several large models; without the additional load. The chapter also identified several areas for improvement, for example the performance with rare classes (like I-MISC or low frequency PoS tags), and discussed possible contributions of the adversarial setup toward generalization overall.

CHAPTER 5

CONCLUSION AND FUTURE SCOPE

5.1. CONCLUSION

The proposed light-weight sequence labeling framework shows that there are high-performance NLP tasks such as Named Entity Recognition (NER), Chunking, and Part-of-Speech (PoS) tagging are contingent upon resource-efficient models like TinyBERT with a Softmax-based sequence-labeling head. By using a lightweight architecture that replaces the software-intensive CRF layers with a light-weight yet effective strategy, it allowed for speed/accuracy tradeoffs across a wider range of computational resources.

Experimental evaluations on CoNLL-2003, CoNLL-2000 and the Brown Corpus showed comparable results in F1-scores, precision, and recall and showed strong generalization capabilities across different kinds of token classification tasks. Moreover, the architecture considerably decreased latency and appears suitable for deployment in realtime applications, particularly in edge scenarios where computing capabilities are limited.

This work represents a useful step towards the formulation of scalable and flexible NLP systems that can sustain good performance without the costs associated with large-sized transformer architectures or complicated decoding methods.

5.2. FUTURE SCOPE

- **Support for Multilingual Datasets:**
 - By extending the framework for token classification to multiple languages through multilingual TinyBERT variants, it could increase the extraction string into more globally applicable tasks against human language or in a cross-lingual context.
- **Integration of Lightweight Attention Enhancements:**

- Incorporating advanced but efficient attention mechanisms like Linformer or Performer could further reduce computational load while improving sequence modeling.
- **Model Quantization and ONNX Deployment:**
 - Using quantized modalities and the export of the model architecture to ONNX may enable deployment fit to smaller power devices (mobile phones or microcontrollers).
- **Incorporation into End-User Applications:**
 - Embedding the model into real-time tools such as grammar checkers, intelligent writing assistants, or chatbots would enhance its practical utility and user impact.
- **Dynamic Label Adaptation:**
 - Engage in research of models that dynamically adapt label sets would allow the architecture to support varying annotation scheme in domains without needing to be fine-tuned and retrained each time.

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CONFERENCE OR JOURNAL DETAILS

We've Submitted our project to **IEEE 10th International Conference on Research in Intelligent Computing in Engineering (RICE-2025)**

which is a conference focused on Artificial Intelligence and Machine Learning and is scheduled on 19th and 20th of December 2025.

Submission Summary

Conference Name

10th International Conference on Research in Intelligent Computing in Engineering

Track Name

RICE2025

Paper ID

153

Paper Title

Sequence Labelling Using Generative Adversarial Networks

Abstract

As the complexity of natural language processing tasks can reveal the shortcomings of classical sequence labeling models which often rely on large data and enormous computational costs, this project introduces a compact and efficient approach for Sequence Labeling using Generative Adversarial Networks (GANs) that improves the accuracy and robustness of label predictions in real-world NLP scenarios. The system employs a TinyBERT-based generator which assigns labels, and a token-level discriminator that provides adversarial feedback so that labels lead to realistic and disentangled sequences.

This framework also employs soft label smoothing and dual-stream training to enhance supervised and adversarial learning when applied in a low-resource context. This architecture supports core sequence labeling tasks such as Named Entity Recognition (NER), Part-of-Speech tagging, and Chunking seven English benchmark datasets (CoNLL-2003, CoNLL-2000, Brown Corpus).

The proposed approach addresses three overarching challenges of existing sequence labeling systems data inefficiencies, prediction inconsistencies, and limited scalable approaches for operating in a low resource context.

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